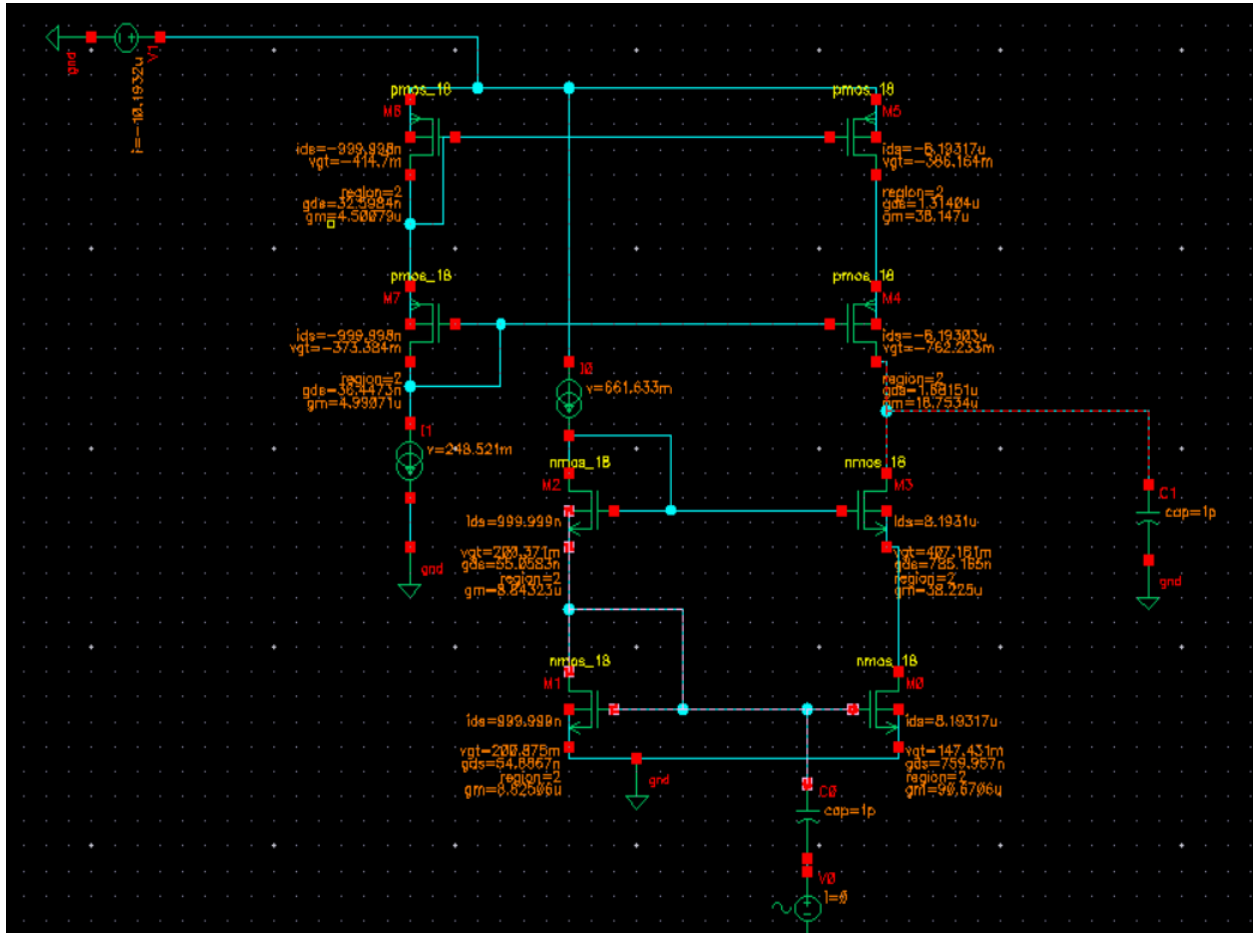


Analog CMOS IC Design
(EE651) Assignment- 4
Prem Singh 25210098

Task 1:- Schematic of a cascode current mirror with active load



Power consumption = $1.8 \times 10.8 = 19.44 \mu\text{W}$
Gain dB = 57.924

ADE L (1) - assign_4 cascode schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

Choose Design

Design Variables

Name	Value
1 L0	1.1
2 L1	2.2
3 L2	2.19
4 L3	6.8
5 L4	3.55
6 L5	1
7 L6	1.5
8 L7	1.2
9 W0	602m
10 W1	420m
11 W2	420m
12 W3	5
13 W4	300m
14 W5	6.72
15 W6	300m
16 W7	300m

> Results in /home/prem/simulation/cascode/spectre/schematic

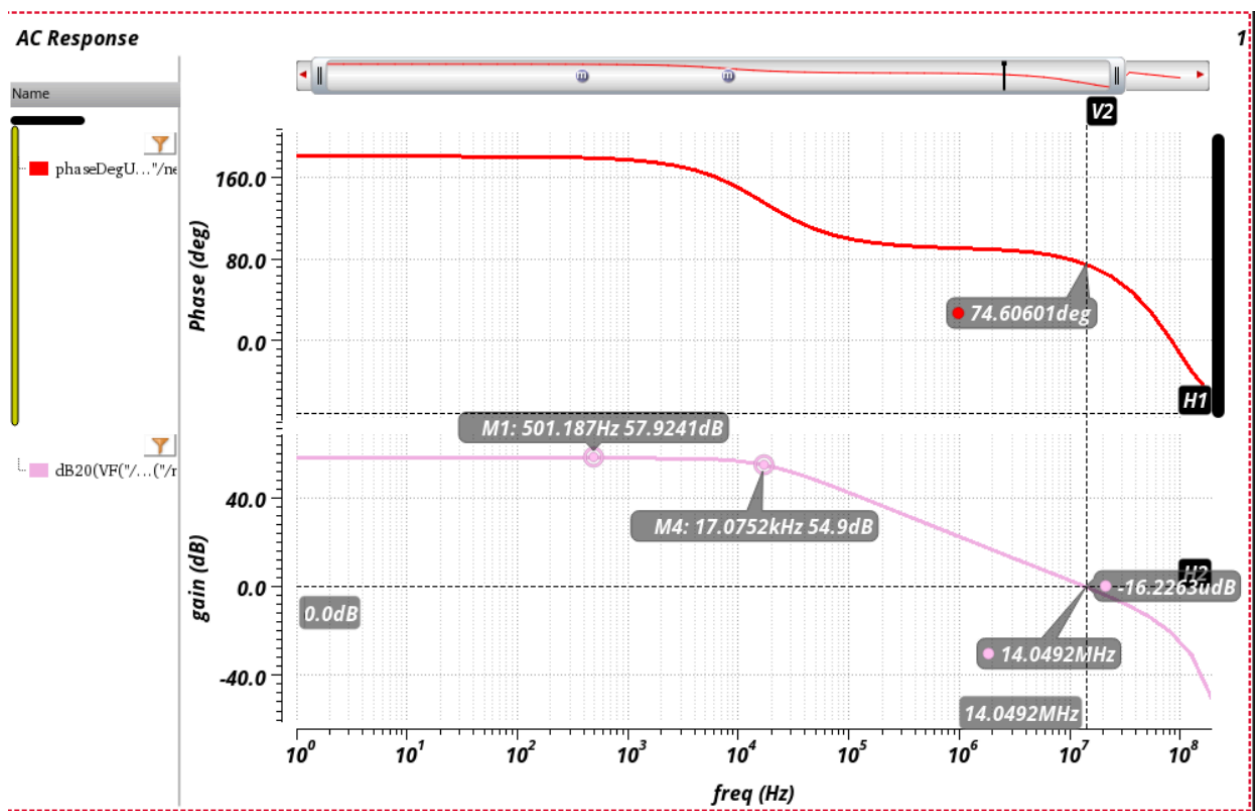
Analyses

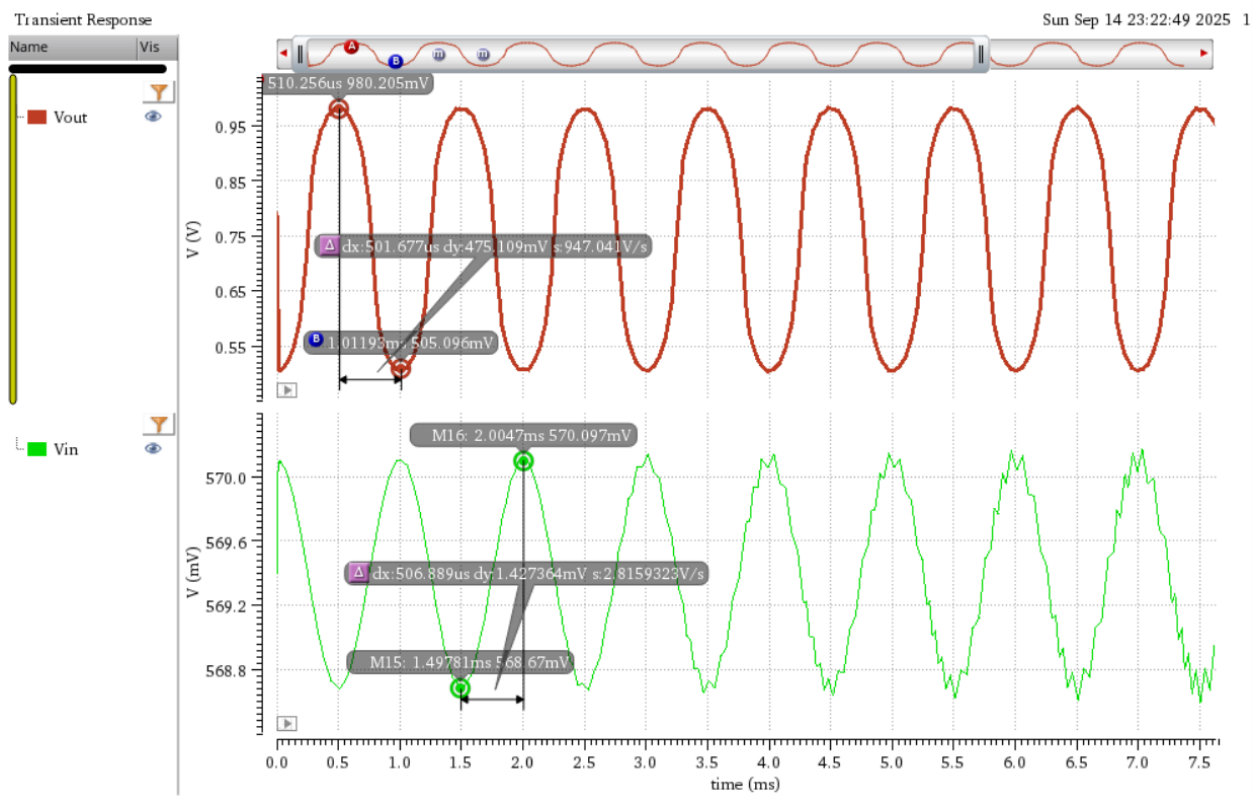
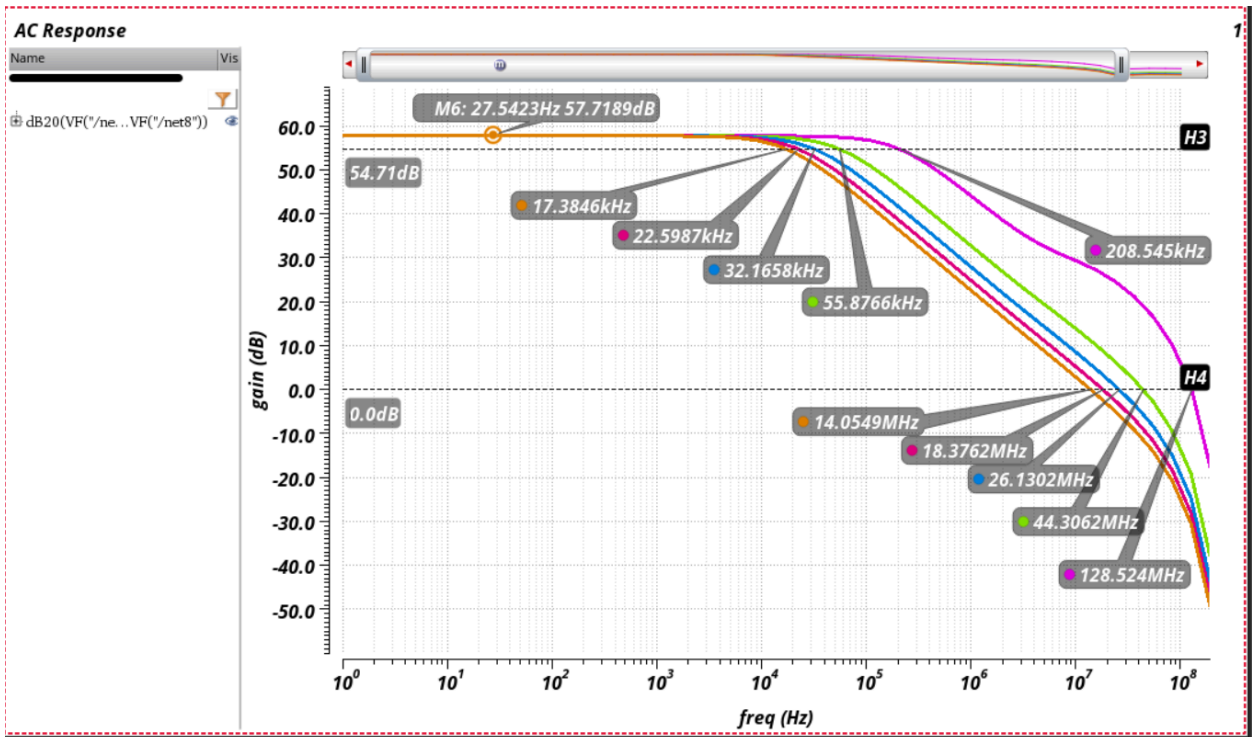
Type	Enable	Arguments
1 dc	☒	t
2 ac	☐	1 1G Automatic Start-Stop
3 tran	☐	0 10m

Outputs

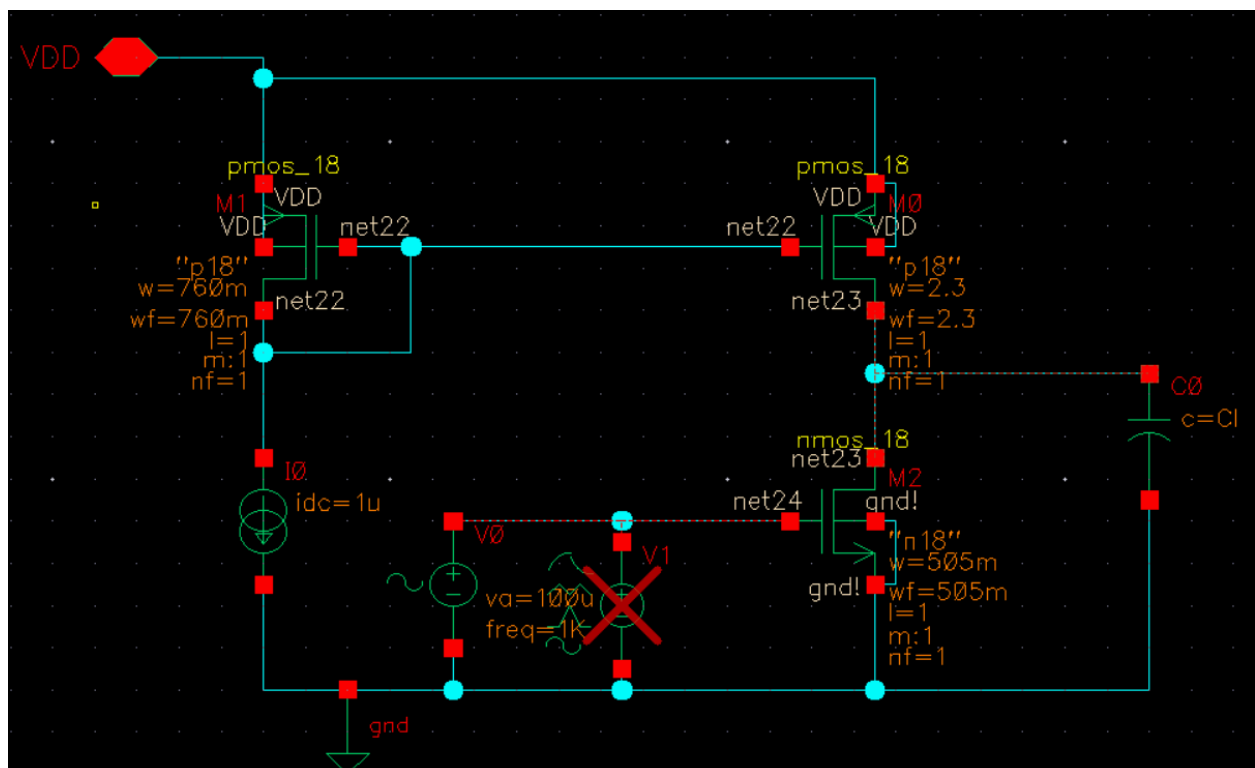
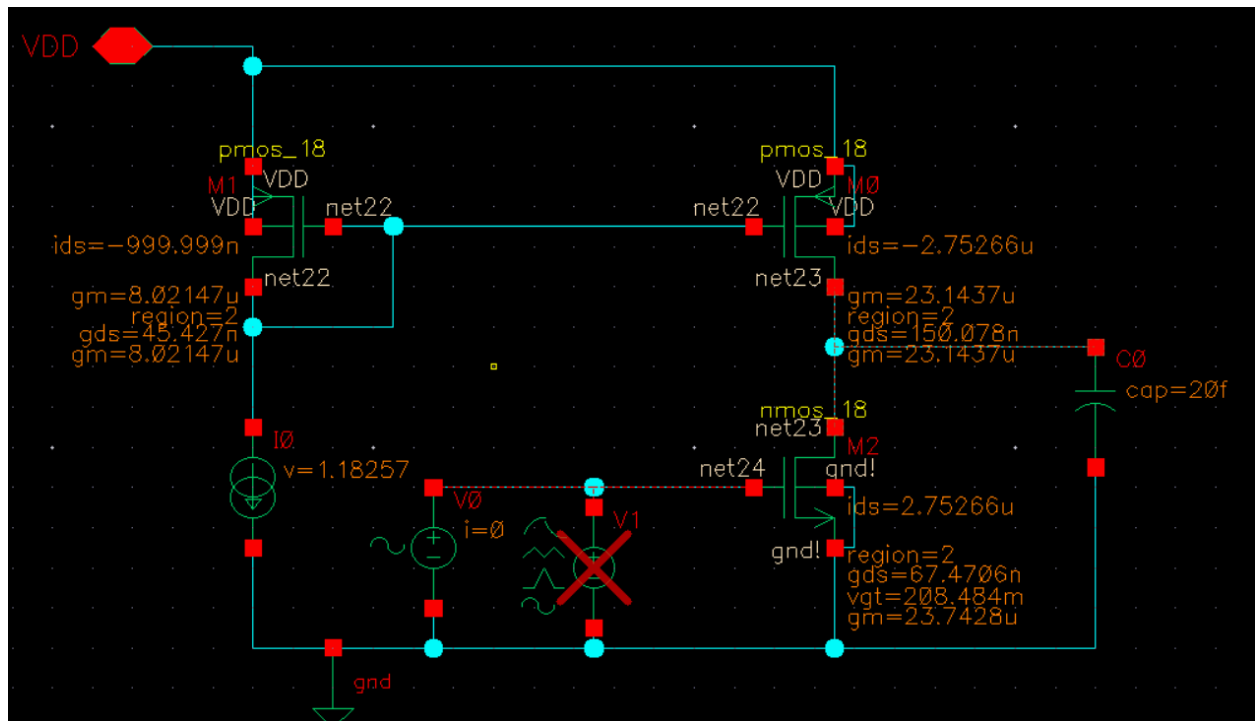
Name/Signal/Expr	/alu	Plot	Save	Save Options
1 net14		☒	☐	allv
2 net8		☒	☐	allv

Plot after simulation: Auto Plotting mode: Ne



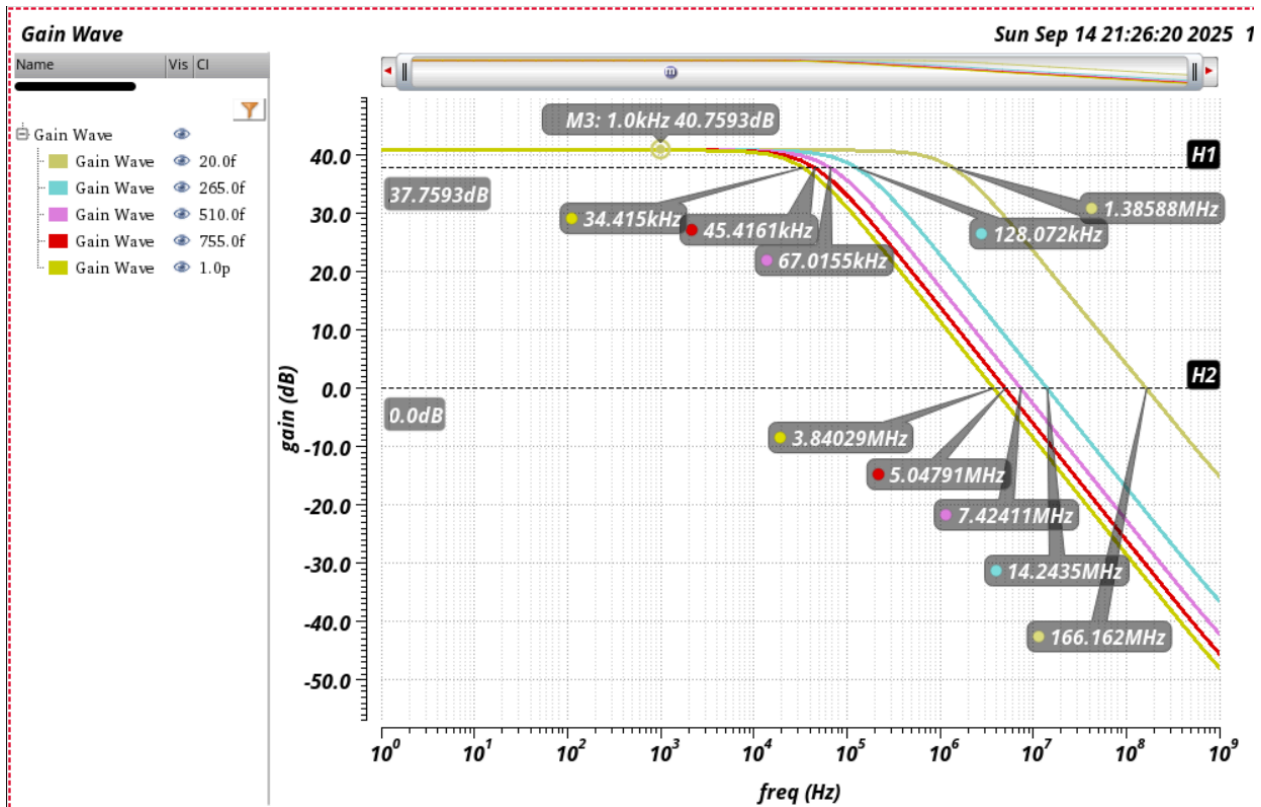


Task 2:- Schematic of a common source amplifier with active load



Power consumption = $1.8 \times 3.7 = 6.66uW$

gain(dB) = 40.7 dB



SN NO:-	Gain (3dB) Task1(KHz)	Gain (3dB) Task2 (KHz)	UBW Task:- 1(MHz)	UBW Task:- 2(MHz)
1	17.3846	34.415	14.0549	3.84
	22.59	45.416	18.376	5.04
	32.165	67.015	26.132	7.424
	55.876	128.07	44.3	14.24
	208.54	1.38MHz	128.52	166.162